

Paraphraser Fingerprinting

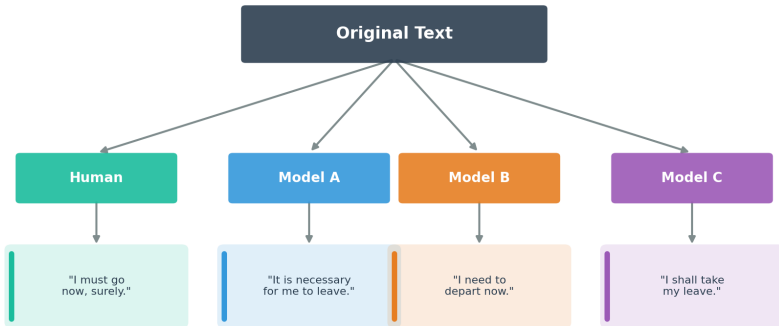
Can We Identify the Rewriter from the Text Alone?

CS6120 Project

Northeastern University

Motivation

Paraphraser Fingerprinting



Same meaning, different style → Can we tell who wrote each?

- Prior work: human vs. machine detection. **Our focus:** *which* model wrote it?
- Applications: AI text detection, academic integrity, model provenance

Two Evaluation Paradigms

Task 1: Duplicate Detection

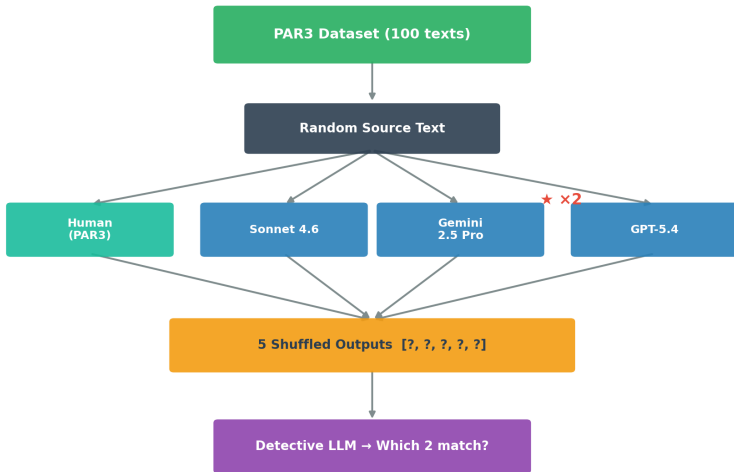
- 4 paraphraser, 1 source text
- One paraphraser writes **twice**
- 5 shuffled outputs
- Detective: “Which 2 are from the same model?”
- Baseline: $\frac{1}{\binom{5}{2}} = 10\%$

Task 2: Pairwise Classification

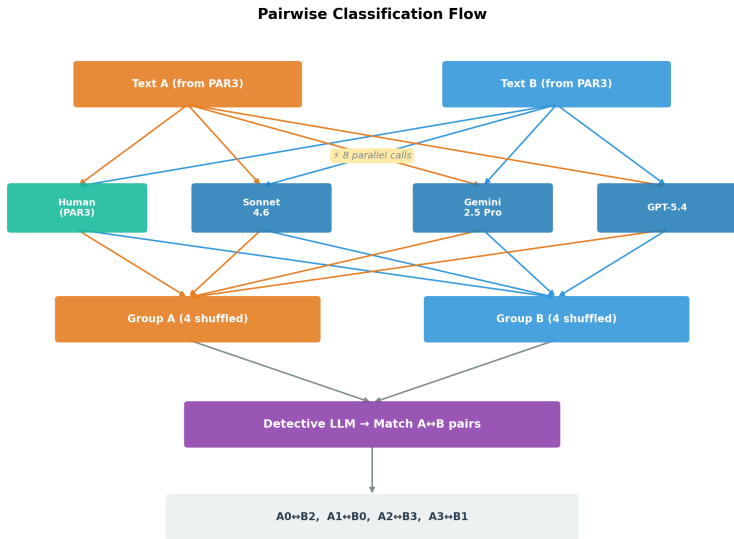
- 4 paraphraser, **2 source texts**
- Each model rewrites **both** texts
- 4+4 shuffled outputs
- Detective: “Match pairs across groups”
- Baseline: $\frac{1}{4!} = 4.17\%$

System Architecture — Duplicate Detection

Duplicate Detection Flow

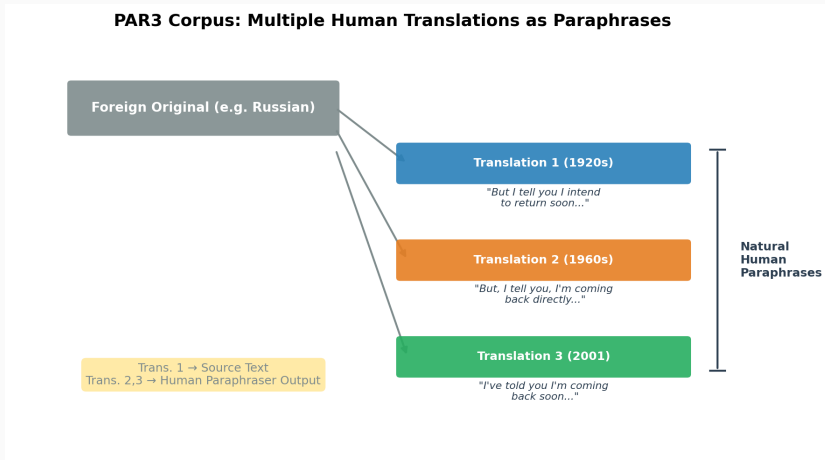


System Architecture — Pairwise Classification



All 4 correct → ✓
Baseline: 1/24 = 4.17%

Dataset: PAR3 Corpus



- Multiple **human translations** of same paragraph $\Rightarrow$ natural paraphrases
- Sources: Dostoevsky, Voltaire, Verne, ... Filter: ≥ 3 translations, ≥ 100 chars

Paraphraser Configurations

“Weak” Set

- Human (PAR3 translations)
- Llama-3.1-8B
- Gemini-2.5-Flash
- DIPPER (T5-based)

Small/specialized models + human

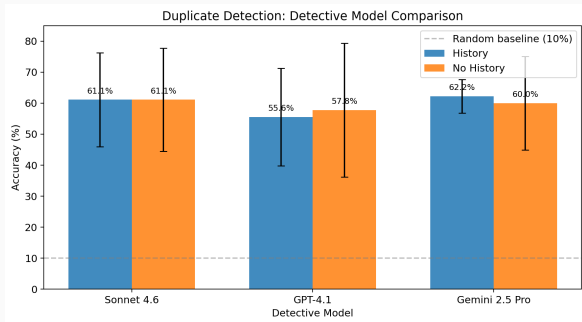
“Strong” Set

- Human (PAR3 translations)
- Claude Sonnet 4.6
- Gemini 2.5 Pro
- GPT-5.4

Frontier models + human

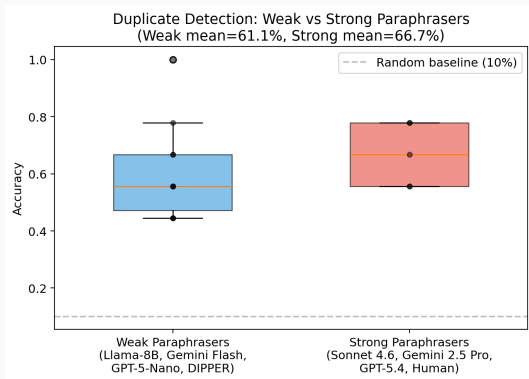
Detective models tested: Sonnet 4.6, GPT-4.1, Gemini 2.5 Pro

Result 1: Duplicate Detection — Detective Comparison



- All detectives: **55–62%**
(baseline 10%)
- **History does not help**
Sonnet: 61.1% ↔ 61.1%
- Gemini 2.5 Pro most consistent
(SD = 5.4%)
- 40+ runs, 400+ scored trials

Result 2: Weak vs. Strong Paraphraser

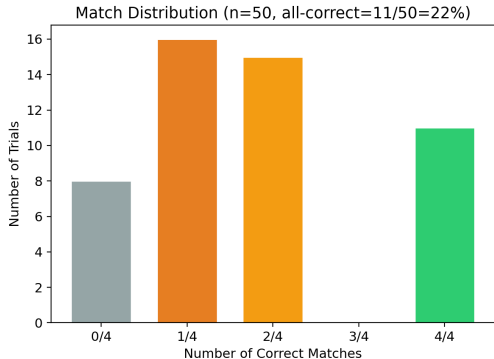
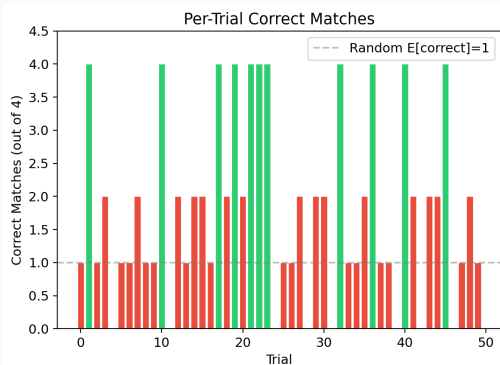


Surprise: Stronger models are *easier* to detect!

Set	Mean Acc.
Weak	61.1%
Strong	66.7%

Why? Capable models develop more consistent “voices” — stronger fingerprints

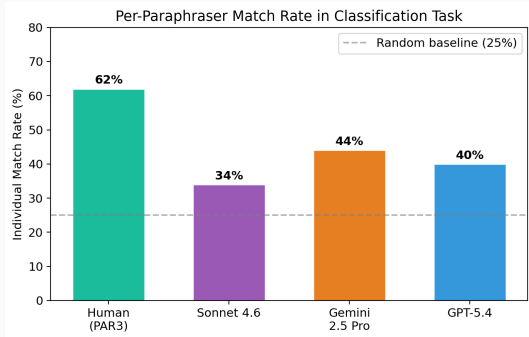
Result 3: Pairwise Classification — Main Result



- 22% perfect match (11/50)
- Baseline: 4.17% $\Rightarrow$ 5.3 $\times$

- **Bimodal:** 0 trials with 3/4
- Either all-correct or multiple errors

Result 4: Who Is Most Identifiable?

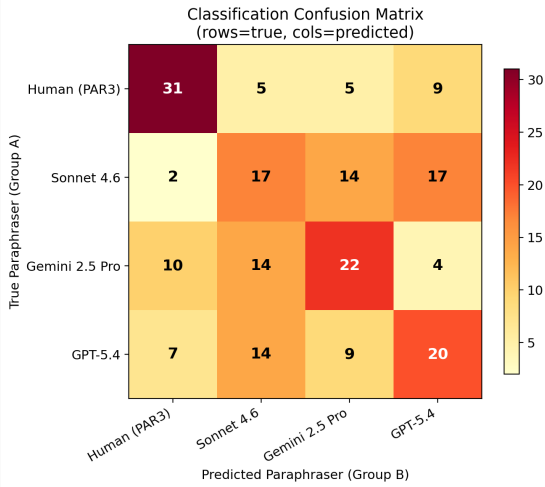


Paraphraser	Match Rate
Human (PAR3)	62%
Gemini 2.5 Pro	44%
GPT-5.4	40%
Sonnet 4.6	34%
Random	25%

Key finding:

Humans are the *most* identifiable!
Sonnet 4.6 is the best “disguiser”

Result 5: Confusion Matrix



- Human rarely confused with LLMs
- LLMs frequently confused with each other
- Human fingerprint occupies a **distinct region** of style space
- Sonnet 4.6 most “generic” — hardest to pin down

Experimental Summary

Summary of All Experiments

Mode	Paraphrasers	Detective	History	Trials	Runs	Accuracy
Duplicate Detection	Weak	Sonnet 4.6	History	10	10	61.1%
Duplicate Detection	Weak	Sonnet 4.6	No History	10	10	61.1%
Duplicate Detection	Weak	GPT-4.1	History	10	5	55.6%
Duplicate Detection	Weak	GPT-4.1	No History	10	5	57.8%
Duplicate Detection	Weak	Gemini 2.5 Pro	History	10	5	62.2%
Duplicate Detection	Weak	Gemini 2.5 Pro	No History	10	5	60.0%
Duplicate Detection	Strong	Sonnet 4.6	No History	10	5	66.7%
Classification	Strong	Sonnet 4.6	No History	50	1	22.0%

8 configurations · 50+ runs · 500+ scored trials

Key Takeaways

1. **Fingerprints are real.**

All conditions far exceed random baselines ($6\times$ for duplicate, $5.3\times$ for classification)

2. **Humans are most identifiable.**

PAR3 human translations: 62% match rate vs. 34–44% for LLMs

3. **Stronger models = stronger fingerprints.**

Frontier LLMs develop more distinctive “voices” (66.7% vs. 61.1%)

4. **History doesn't help.**

Within-trial stylistic comparison matters more than cross-trial memory

5. **Pairwise classification is hard but feasible.**

22% perfect match with bimodal success pattern

Future Work

- **Scale paraphraser pool:** 8+ models, same-family comparison (e.g., Llama-8B vs. Llama-70B)
- **Feature-based classifier:** Complement LLM detective with traditional NLP features (n-grams, syntax)
- **Robustness:** Do fingerprints survive post-processing (grammar correction, back-translation)?
- **Cross-domain:** Test across scientific, literary, conversational text
- **Embedding analysis:** Use embedding similarity as a quantitative fingerprint metric

Thank You

Questions?

Code: `https://github.com/zboyrr/HideNSeek`